

Boundary-Relative Temporal Experience and Judgment: Process History, Represented Extent, and Query-Specific Readouts

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Abstract

Temporal experience and judgment require distinctions among processing history, represented temporal extent, and query-specific information use. This article proposes a systems methodology that separates modeled dynamics, registration, functional access, represented content, and observable responses while distinguishing clock observation, duration calibration, and scheduled action. Three inferential levels are kept distinct: conceptual object typing (Level 1); prospective admissibility rules based on information lineage, query referent, constrained transformation, and boundary lineage (Level 2); and registered empirical implementations (Level 3). Test A separates D1a, support for a positive represented-extent reconstruction pattern, from D1b, incremental predictive utility over same-information unrestricted and matched alternatives. Practical-null, directional, opposite-sign, and indeterminate regions are predeclared. Test B independently evaluates measurement preservation and interface sufficiency across enlarged analytical boundaries. The framework specifies prospective reusability as a criterion to be tested, not as an established result.

Keywords: subjective time; temporal judgment; system boundaries; process history; represented duration; event memory; model restriction

Practitioner Points

- Measure felt passage, judged elapsed duration, and scheduled action separately; a change in one does not establish improvement in the others.
- Preserve how a temporal measure was defined when enlarging a model to include clocks, reminders, or other people.
- Treat a person's report of extensive temporal content as a report requiring interpretation, rather than a direct count of experienced transitions.

1. Introduction and scope

Two people can share an external interval without reporting the same temporal experience. More consequentially, one person can describe an interval as slow while it unfolds, brief in recollection, and rich in represented years. These judgments have different referents. A systems account should specify which record supports each judgment, how external information reaches the system, and what remains comparable when an analyst changes the system boundary. The central proposal is a dissociation architecture. Processing history, represented temporal content, duration estimation, passage reports, and clock-linked action are partly separable objects and functions. The framework is not a rigid wiring diagram: it defines a restricted space of admissible functional relations among typed variables, while allowing distributed, overlapping, history-dependent implementation. Crucially, the restrictions are not licensed merely because a coefficient pattern appears sensible for a particular experiment. Before confirmatory outcomes are examined, the analyst must declare the information lineage of each modeled feature, the referent of each query, any transformation required to make one usable for the other, and the boundary under which those declarations are made. A transformation name alone is insufficient: a cross-referent route must also specify its input type, target referent, operational or independently motivated theoretical basis, and a discriminating empirical or measurement consequence. Experiment-specific statistical constraints instantiate those prior declarations. The framework's empirical value therefore depends on whether this prospective discipline can generate predictive or transfer structure on tasks not used to formulate a particular implementation, beyond established attention, memory, event-structure, semantic-inference, and monitoring accounts. The assertion that reports depend on internal processing supplies motivation, but does not itself establish a theory.

The scope is adult human supra-second episodes, especially tens of seconds to minutes, and subsequent judgments about those episodes. Outputs include felt passage, estimated laboratory duration, remembered episode duration, represented narrative extent, and scheduled action. Order and detail tests are auxiliary measurements of the records supporting those outputs. No claim is made to explain all temporal psychophysics. Intentional binding and cardiac

effects on subsecond interval judgments establish relevant limits [17,18], rather than phenomena to be absorbed into an unrestricted readout function. Extending this account to such tasks would require a new specified implementation. The framework is conceptually compatible with Boundary-Relative State-Time: Measurement Lineage Across Nested System Boundaries [1]. That approach requires an enlarged analytical description to preserve inherited history records and measurement semantics when resolution is retained. Here that discipline guards comparisons across cognitive and person-plus-clock descriptions. Subjective reports are not identified with the companion framework's additive scalar state-time.

Remembered duration can be nonadditive, reconstruction can change later reports, and neither entails a violation of physical history preservation. The contribution is a systems methodology for preserving interpretability and testing transfer across analytical descriptions, not a new mathematical invariance principle, regression family, or universal causal law of temporal cognition. Its narrower commitments are fourfold: a Level 1 typed distinction between process and represented content; a Level 2 pre-experimental admissibility procedure linking information lineage, query referent, operationally constrained transformations, and boundary lineage; Level 3 explicit statistical restrictions that instantiate those rules for registered tasks; and a separation between an analytical preservation rule and an empirical claim about the sufficiency of an interface record. Test A bears the primary empirical burden for the within-boundary temporal program but separates two inferential decisions: D1a tests the registered reconstruction-pattern hypothesis, whereas D1b tests whether imposing that pattern earns incremental predictive utility. Either can be reported without being conflated with the other. Successful results can support the usefulness of the broader relational methodology only if its restrictions were prospectively generated and later transfer; neither result turns the underlying object-type distinction into an empirical discovery. Test B/D2 bears a separate empirical burden for the cross-boundary systems claim. Neither Test A nor Test B substitutes for failure of the other.

2. Theoretical method and closest alternatives

This is a conceptual analysis with proposed experiments, not an empirical report or systematic review. Literature was selected to challenge the distinctions claimed here: contextual change and event memory, perceptual-state accumulation, compressed recollection, semantic duration, episodic simulation, clock monitoring, and bodily timing. No participant data, effect estimates, or model-fitting results are reported. Definitions, design choices, and illustrative mathematical constraints are proposals unless explicitly attributed to a source.

2.1 Timing and event-memory accounts

Prospective versus retrospective timing concerns whether the participant knows in advance that duration will be judged [2,3]. Online versus later reporting concerns when a response is made. These axes must not be equated: a prospective duration estimate can be collected after an interval, and a passage rating is not the same measure as either duration task. The established dissociation motivates multiple outputs but does not make query indexing novel. Contextual-change accounts and segmentation experiments already connect encoded changes with duration judgments [4]. Neural pattern change during narrative encoding predicts retrospective interval estimates [6]. Learned episodic structure can also influence prospective judgments [7], so the revision does not assign segmentation exclusively to retrospective timing. Narrative temporal shifts themselves can be perceived as event boundaries and can alter processing and memory [23]. Recent work further shows that internal event structure can systematically distort subjective duration [24], and that remembered subevent number predicts retrospective duration judgments of naturalistic whole events [25]. These findings strengthen event structure as a direct competitor rather than a nuisance covariate.

Zacks et al. [5] link event-model updating to prediction failure, while the more recent synthesis by Stojić and Nadasdy [20] puts events at the center of psychological time. Thus, endogenous events, segmentation, and model updating cannot serve as novelty claims here. An event-memory model allowed to retain distinct kinds of content may already capture the proposed dissociations. Roseboom et al. [9] offer a particularly close computational precedent: changes in perceptual classification activity are accumulated and mapped to duration labels, with gaze-constrained input improving correspondence to human estimates. This already joins selected state change with external-duration mapping. State-dependent neural timing [10] supplies another implementation family. Neither the absence of a single clock nor a learned conversion to seconds distinguishes the present proposal from these accounts.

2.2 Represented duration and simulation

The distinction between processing duration and represented duration has direct antecedents. Jeunehomme and D'Argembeau [11] compared event duration with the time required to mentally replay real-life episodes; their findings relate compression and judged duration to recalled moments. Work on event duration in language shows that semantic and contextual information supports estimates of how long described events last [12]. Episodic construction studies also distinguish the construction of remembered and imagined events [13]. Accordingly, neither compressed history nor the ability to represent years within seconds is new. The present claim does not require T^* to contain empirical information unavailable to a semantic model. In Test A, T^* may be the same independently calibrated semantic/temporal-extent feature supplied to M2 and M3. The stronger question is whether assigning that observable a represented-content role, and constraining its relations to different queries accordingly, yields a cross-query pattern and held-out predictive advantage beyond event structure, memory detail, semantic anchoring, and task demand.

Merely asking how many fictional years a label denotes cannot answer that question.

2.3 External clocks, embodiment, and inference

Clock monitoring and time-based prospective memory already have explicit functional models. Munaretto et al. [14] relate changing monitoring patterns to performance near a deadline. Laera et al. [15] manipulated clock speed and found monitoring changes without an effect on prospective-memory performance. These are close precedents for separating clock use from other temporal measures, not evidence unique to Boundary Theory. Interoceptive processes may supply encoded information or alter processing. Wittmann [16] develops a bodily account of temporal experience; cardiac-phase experiments [17] show why bodily influence cannot simply be equated with consciously identifying a heartbeat. In this architecture, physical bodily variables enter modeled dynamics; their encoded consequences enter the registered record. If a bodily variable is instead treated as a gain modulator, that role and its measurements must be specified in advance. It cannot be used interchangeably as input, gate, and explanation of residual error.

Prediction-error accounts offer candidate mechanisms for updating registered records [5]. They do not establish a unique implementation of the present architecture. Semantic priors and causal interpretation are also serious alternatives, particularly for narrative experiments. Intentional binding [18] shows that judged timing can depend on action structure; the current scope restriction prevents that family from being explained merely by adding an unnamed input to the readout.

3. Typed architecture

3.1 Boundary, dynamics, and registration

Let B be a declared analytical system boundary selected before outcome analysis. For a person-level model, specify sensory inputs, bodily variables, task states, memory variables, and output channels at a fixed sampling resolution. B is not defined by whether a variable happens to correlate with the eventual report, and is not claimed to locate a phenomenal or anatomical border of consciousness. Let $x_B(t)$ be the modeled internal state and $\Gamma_B[t_0, t_1]$ its process trajectory. External laboratory time t indexes the record; $\Delta t = t_1 - t_0$ is the measured interval. Γ_B is an analyst's representation of modeled dynamics, not a record assumed to be wholly stored or consciously inspected by the participant. The earlier notation Γ_C is retained only when C names a declared cognitive subsystem under this same rule, rather than a boundary defined by consciousness.

Registration is the encoding process $E_B = G_B(\Gamma_B; g)$, where g declares detector variables, sampling, and update rules. The encoded record E_B can include implicit information, perceptual states, bodily effects, and memory traces. A physical change need not be preserved in E_B . Its occurrence, its registration, and its later availability are distinct hypotheses.

3.2 Functional access and conscious report

At query time u , $A_q(u) = F_q(E_B, I_q, u; f_q)$ denotes information usable by the specified temporal computation. I_q contains the query instruction and declared context. F_q can retrieve or transform information; access does not

require that the source itself be consciously identified. A learned expectation may influence duration without appearing as a reportable thought. Phenomenal experience $P(u)$ is not equated with A_q . Reportability is measured through responses such as content discrimination, confidence, or an awareness report; it is an imperfect indicator, not an exhaustive measurement of P . The main empirical output is y_q , the observed temporal response, which can depend on accessible information and reporting conventions. Claims about experience must therefore be stated as interpretations of convergent reports. This architecture neither requires nor licenses an inference that all causes of a conscious duration judgment were conscious contents.

3.3 Process history and represented temporal content

Γ_B is a trajectory through modeled states. $T(u)$ is representational content at a time: a structured description of events, temporal relations, implied duration, and source information. These have different types. T may arise from an encoded episode, semantic inference, an external story, imagination, or reconstruction. A compact representation of a decade is a present state with decade-related content; it does not contain a decade of physical processing.

$$T(u) = J(E_B \leq u, \text{semantic knowledge, narrative input, instruction; } j). \quad (1)$$

J is a dependency schema, not a specified neural algorithm. Representing T also requires processing and therefore contributes to Γ_B . The distinction is between the history of constructing a representation and the temporal interval it describes. They can be causally coupled without sharing units or being proportional. For experiments, define L as the randomized sequential-richness manipulation and L^* as an independently calibrated proxy for task-defined sequential structure and the process-derived record it can support. Neither is a count of phenomenal moments, an operational definition of Γ_B , or a scalar amount of total process history. No universal monotonic $L^* \rightarrow \text{duration}$ rule is assumed. Define T^* as the independently calibrated semantic/temporal-extent feature of represented content, such as the

duration implied by a scenario and its accepted temporal relations. T^* is not proposed as a new latent quantity: the same observable is supplied to $M2$ and $M3$. The theoretical claim concerns its typed functional role and the restrictions that role imposes across queries. T^* must not be obtained from the same target response it is used to predict.

3.4 Encoded structure versus observed answers

O_E denotes encoded precedence constraints, possibly partial or inconsistent. S_E denotes encoded segmentation at the declared event scale. Reported order y_{order} and boundary marks y_{segment} are noisy measurements, influenced by retrieval and response demands. They are not identical to O_E and S_E and cannot be simultaneously treated as latent inputs and target outcomes without an explicit measurement model. Inconsistent order reports should be analyzed as uncertainty or error rather than converted post hoc into a different internal ordering rule. Broad free parameters M and V are removed. Experiments may include narrowly measured variables such as imagery vividness, self-involvement, arousal, or task importance, but must define their measurement and role before testing. Unmeasured “mode” or “value” cannot be added to rescue a failed prediction.

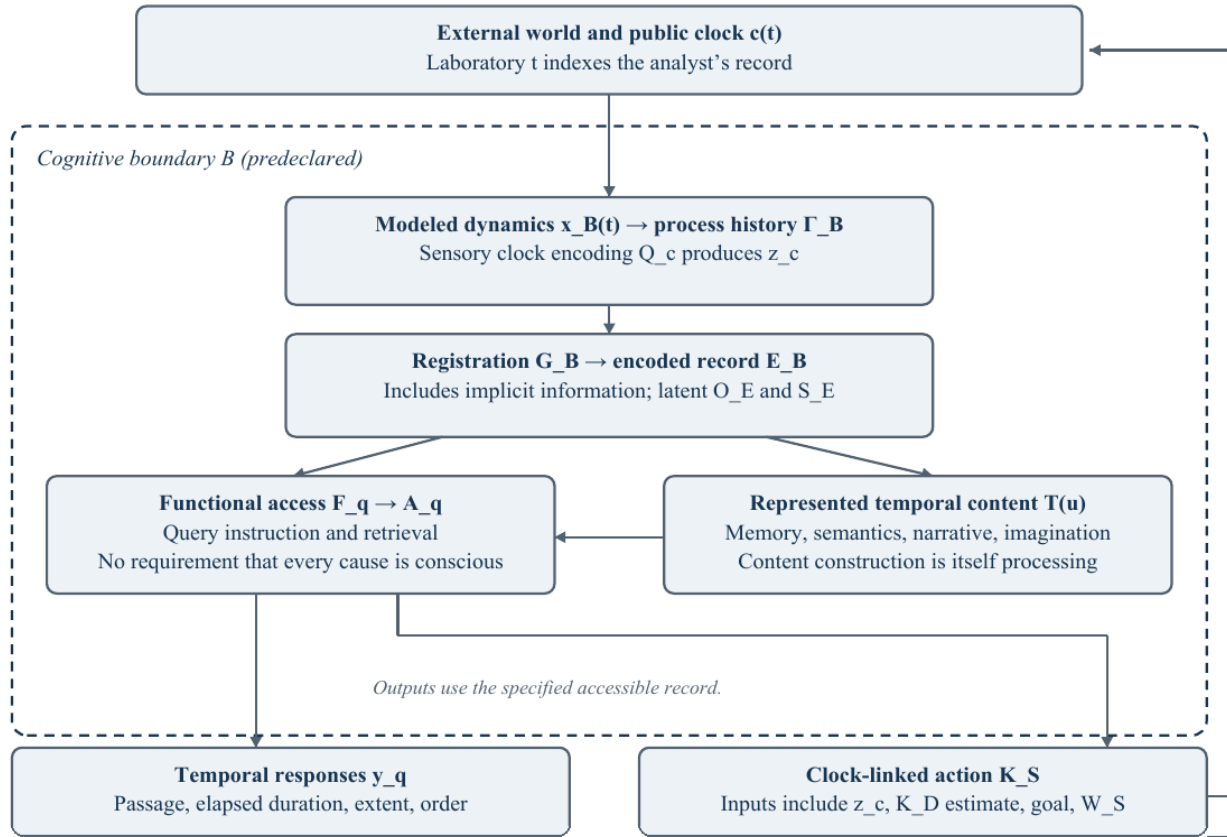


Figure 1. Functional architecture. Arrows specify candidate information pathways, not a requirement that processing be serial or conscious. Clock-linked actions feed back to the environment. Phenomenal experience is not inferred directly from any box.

Notation and construct guide

Symbol / label	Role in this manuscript
B	Predeclared analytical system boundary.
t, Δt	External/laboratory clock coordinate and measured interval.
$x_B(t)$, Γ_B	Modeled internal state and its process trajectory/history.
E_B	Encoded record produced by registration G_B ; may contain implicit information.
A_q	Information functionally usable for query q; not equated with phenomenal experience.
$T(u)$	Represented temporal content at query time u; distinct in type from process history.
O_E , S_E	Encoded ordering/precedence and segmentation structure.
L, T	Randomized Test A conditions: sequential richness and represented extent; T is shorthand distinct from $T(u)$.
L^* , T^*	Independently calibrated observables for sequential structure and represented temporal extent; T^* is the same semantic/extent feature supplied to M2 and M3, not a new latent quantity.
Q_c , z_c	Clock encoding function and available clock representation.

Symbol / label	Role in this manuscript
K_D	Unaided duration-calibration function.
K_S, W_S	Scheduled-action policy and predeclared weighting of the clock-linked goal.
y_q	Generic observed temporal response for query q.
Y, Y_order, Y_extent	Test A personal-duration, serial-order, and represented-extent response channels.
h, Z	Fixed internal-history proxies and declared clock/interface record used in Test B transfer.
M0-M3	Reduced-model hierarchy; M2 is the unrestricted same-information competitor. M3-R is the proposed represented-extent reconstruction model; matched M3-N/M3-O/M3-P alternatives preserve the direct validation routes and alter only the theory-relevant duration-routing restriction.
D1a, D1b, D2	D1a tests the registered within-boundary reconstruction pattern; D1b tests incremental predictive utility of imposing that routing; D2 separately tests cross-boundary interface sufficiency.

This guide is descriptive only; operational definitions and restrictions remain those specified in the relevant sections and registered test implementations.

3.5 Clock observation, calibration, and synchronization

Let $c(t)$ be an environmental clock display. Q_c is sensory and symbolic encoding; $z_c(u) = Q_c(c(t)$, observation history) is the participant's available clock representation. Reading z_c is clock observation. Inferring elapsed laboratory time without an observed clock requires a different function, K_D , fitted from internal-record features and training experience. Public t does not enter a participant's decision as an unmediated input.

$$\hat{d} = K_D(A_{\text{duration}}, \text{training context}; \theta_D). \quad (2)$$

K_S is a policy for acting on a schedule. It uses observed clock information when available, internally estimated time, a maintained goal, and declared action costs. W_S is a separately specified weighting rule for the priority of a clock-linked goal relative to an ongoing task. W_S is not another unrestricted cause of felt duration.

$$a(u) = K_S[z_c(u), \hat{d}(u), \text{goal}(u), W_S; \theta_S]. \quad (3)$$

The experimental outcomes differ: error of an unaided estimate assesses K_D ; clock-reading accuracy assesses Q_c ; check frequency and action timing constrain K_S ; controlled incentive manipulations constrain W_S . They may interact. Functional separability predicts selective intervention and transfer patterns, not unconditional statistical independence. Durable recalibration requires evidence of changed unaided performance on later trials; responding to a newly visible clock is insufficient.

4. Restricted readouts and measurement commitments

$$y_q \sim p_q(y \mid \phi_q[A_q], \theta_q). \quad (4)$$

Equation (4) is an architectural dependency relation. ϕ_q is a fixed feature map for a specified query, and θ_q belongs to a predeclared model family. It is not a quantitative law until features, scales, link functions, and estimation procedures are supplied. The architecture does not require hard one-to-one mappings between variables and outcomes. Instead, declared object types constrain the hypothesis space, and the rules below determine how those types can be related to a query before experiment-specific coefficients are chosen. These are systems-model restrictions, not claims of serial brain modules or statistical independence. A study must fix the declarations and their statistical instantiation before confirmatory data are examined. The word “accessible” cannot be defined merely as “whatever predicts y_q .”

4.1 Pre-experimental admissibility rules

Level 2 has scientific content only if its routing classes can be generated before the confirmatory outcomes of a task are known. The framework therefore uses four declarations as a prospective admissibility procedure. They constrain interpretation without requiring deterministic one-to-one behavior. Rule 1 - information lineage and type. Every modeled feature must be assigned a declared informational lineage at the chosen boundary: for example, process-derived or encoded sequential structure, represented content, an external-interface representation, learned calibration, or a goal/policy variable. The declaration states what information the feature is about and how it entered the modeled system; it does not assert a unique anatomical location or biological module. Rule 2 - query referent. Every query q must declare the object or interval about which its response is requested. A predictor-query relation is referent-direct when the predictor contains information of a type that can answer that referent without an additional referent-changing transformation. Thus accessible order evidence is a referent-direct source for an order query, represented temporal extent is a referent-direct source for a represented-extent query, and a mediated clock representation z_c is a referent-direct source for a clock-reading query. “Referent-direct” classifies the relation between information and query referent; it does not imply a universal positive coefficient, a large effect, statistical independence from other sources, mandatory use by the participant, or a rigid neural route.

Rule 3 - declared transformation for cross-referent use. Information may influence a query whose referent it does not directly answer, but a cross-referent route is admissible at Level 2 only through a transformation specified before confirmatory outcomes are examined. Naming “reconstruction,” “inference,” “calibration,” “retrieval,” or “action policy” is not by itself sufficient. The declaration must identify (a) the input information lineage/type, (b) the target query referent, (c) the operation and its operational interpretation or independently motivated theoretical basis, and (d) at least one discriminating empirical or measurement consequence that differs from leaving the cross-route as an unconstrained free coefficient. Such a consequence may be a registered sign/region, required intermediary measurement, selective transfer prediction, or another predeclared restriction. If these elements cannot be specified prospectively, the coefficient may still be explored, but it is not licensed as evidence for the typed relational architecture. In Test A, the Level 3 reconstruction route satisfies this rule by declaring T^* as input, personal experienced duration as the target referent, retrospective reconstruction as the proposed operation, and the registered positive region with matched null and opposite-sign alternatives as its discriminating consequence; a null, opposite, indeterminate, or noncompetitive result cannot be reinterpreted as support.

Rule 4 - boundary-lineage preservation. When the analytical boundary is enlarged, inherited type labels, records, query referents, and measurement specifications must remain recoverable at retained resolution. New variables may open additional routes, but they cannot retroactively redefine what an inherited variable measured or what an earlier response referred to. Section 5 formalizes this requirement for the present boundary comparison. Prospective derivation criterion. Before confirmatory analysis, a new task should be representable as a routing table containing at least: source feature and lineage, query referent, whether the relation is referent-direct or cross-referent, any required transformation specification, and the declared boundary. For a cross-referent route, the table must also record the transformation's operational/theoretical basis and its discriminating empirical or measurement consequence. Statistical restrictions are downstream implementations of that table. If the framework cannot classify a new task, or cannot constrain a proposed transformation beyond naming it, without consulting the outcome pattern, Level 2 has added no reusable content beyond experiment-specific preregistration.

This differs from generic domain-informed constrained multivariate modeling in the source and preservation of the constraints, not in the mathematics used to fit them. A conventional model may impose exactly the same coefficients and, if it adopts the same declarations, is empirically equivalent for that task. The claimed systems contribution is therefore methodological: information lineage + query referent + operationally constrained transformation + boundary generate an admissible relation class, and experiment-specific statistical restrictions instantiate that class. A transformation that does not constrain interpretation, measurement, sign/region, transfer, or another discriminating consequence adds no Level 2 content. The statistical model is downstream of that representational discipline rather than the source of it. Prospective illustration on a new task. Suppose one episode is followed by four questions not used to construct Test A: (i) which event occurred first, (ii) how long a character waited within the described scenario, (iii) how long the participant personally felt they were waiting, and (iv) when to press a button at a specified public-clock time. Before fitting any outcome, the rules classify process-derived order evidence as a referent-direct

source for (i), represented temporal extent as a referent-direct source for (ii), and clock-linked action in (iv) as requiring a mediated clock representation plus a declared goal/policy relation rather than direct access to laboratory t. Neither represented story duration nor sequential evidence directly answers (iii); any proposed influence on personal-duration judgment therefore requires a prospectively specified reconstruction/access transformation with declared inputs, target referent, basis, and discriminating consequence. The point is not that these effects must occur, but that the architecture determines what must be declared and made risky before they can be interpreted.

4.2 Levels of claim and inference

Level 1 is the type distinction: a modeled process trajectory Γ_B and represented temporal content T are different kinds of objects. Test A does not establish this distinction as an empirical discovery; it uses the distinction to define what relations are being compared. Level 2 is the relational systems methodology: the pre-experimental rules in Section 4.1 restrict the admissible cross-query relation space without requiring rigid one-to-one mappings. Their proposed reusable content lies in classifying relations from information lineage and query referent, and in requiring cross-referent transformations to be prospectively specified strongly enough to produce a discriminating empirical or measurement consequence. A named transformation with no such consequence does not license a route. Two empirical questions are then kept separate. Coefficient-pattern support asks whether observed relations occupy the registered region of that prospectively derived space; incremental predictive utility asks whether imposing those restrictions improves held-out prediction or predictive economy relative to a same-information unrestricted model. Held-out advantage can support the incremental usefulness of the restrictions, but an unrestricted supermodel may learn the same coefficient pattern with sufficient data. A conventional cognitive model that adopts the same pre-experimental declarations is empirically equivalent for that implementation; the claim is a systems discipline for generating and preserving constraints, not exclusive ownership of their statistical form or a stronger causal theory than the declarations warrant. Prospective reuse across new tasks remains a criterion to be tested rather than a demonstrated property.

Neither result establishes anatomical separability or a unique biological implementation. Level 3 is the registered implementation: for Test A, represented temporal extent T^* is predicted to contribute positively to retrospective personal-duration reconstruction while the direct order and extent relations satisfy their declared constraints. D1a tests the registered reconstruction-pattern hypothesis: whether the direct/cross-channel relations and the positive T^* duration route occupy their preregistered regions and generalize to the held-out relational narrative family. D1b separately tests architectural utility: whether imposing M3-R produces a practically meaningful held-out predictive gain over unrestricted M2 and the best matched restricted alternative without degrading the auxiliary channels beyond declared tolerances. D1a can be supported while D1b is not; in that case the reconstruction pattern is supported but the restriction has not earned incremental predictive economy. Failure of D1a defeats the registered reconstruction implementation. Repeated inability of prospectively generated Level 2 restrictions to add predictive or transfer value defeats their claimed incremental usefulness, while Level 1 remains a conceptual typing distinction rather than a rescued empirical result.

Mandatory roles differ by query. An unaided seconds estimate uses an interval record and K_D . An observed-clock calculation uses z_c and symbolic operations, and is scored separately. A represented-extent response uses T-related features. An order report uses accessible O_E -related evidence. A passage report uses a time-windowed record and its explicit passage instruction. A retrospective episode-duration report uses retrieved features associated with the target episode. Clock-linked action uses equation (3). None is permitted direct access to the analyst's full Γ_B . Optional features are added only when declared for that experiment: contextual diversity, temporal attention, arousal, retrieval strength, and semantic interpretation. The presence of a learned scale for seconds does not imply that passage ratings are computed in seconds. Additive changes in a latent score must not be interpreted as additive lived time. Reports taken at different delays may legitimately differ because retrieval differs.

Table 1. Constructs, proxies, outputs, and alternatives.

Construct	Operational proxy	Target / restriction	Competing explanation
Process Γ_B / L^*	Logged updates; fixed-denominator serial probes	Primary registered relation to order/history-sensitive output; contribution to personal-duration reconstruction is permitted but not fixed in sign by the architecture	Task load; predictability; event boundaries; practice; retrieval burden
Encoded O_E, S_E	Order tests and boundary marks in a separate sample	No target response reused as its own predictor	Retrieval error; explicit marking changes encoding
Represented T / T^*	Independent semantic/temporal-extent calibration; later comprehension	Same observable supplied to M2/M3; primary relation to represented extent and registered positive contribution to personal-duration reconstruction	Semantic reconstruction; event-model updating; numerical anchoring; demand
Functional A_q	Planned retrieval cue; independent retrieval probes	Query-specific influence of encoded content	General attention or memory-strength effect
Clock Q_c / z_c	Display/exposure log; reading accuracy	Direct reading is distinct from unaided estimate	Visibility; reading or arithmetic error
Calibration K_D	Clock-absent estimate and transfer error	Change must persist in unaided trials	Response strategy; regression to mean
Policy K_S / W_S	Clock checks; action latency; incentive condition	Schedule performance, not felt passage	Standard prospective-memory monitoring

No measured boundary count or serial-detail score is a perfect latent-state measure. Use multiple indicators where practical and assess reliability. Analyze randomized total effects before treating post-exposure memory measures as covariates; those measures may mediate an effect and conditioning on them can change the question. Predictive incremental-value analyses are a distinct second analysis. They do not establish mediation or conscious access by themselves.

5. What boundary analysis adds

5.1 Preservation under analytical enlargement

Consider B_1 , the cognitive system with a declared input/output record, and B_2 , that system plus its clock and reminder interface. Enlarging the analysis of one completed run does not physically alter the run. At inherited resolution, a projection π_{21} must recover its B_1 history and a rule restriction ρ_{21} must recover its measurement specification m_1 .

$$\pi_{21}(\Gamma_{B_2}) = \Gamma_{B_1}; \rho_{21}(m_2) = m_1. \quad (5)$$

Consequently, any previously defined statistic $f(\Gamma_{B_1}; m_1)$ remains recoverable as $f(\pi_{21}\Gamma_{B_2}; m_1)$. If an analyst changes the event threshold, access proxy, or seconds mapping when adding a clock variable, that is a model revision. It cannot be reported as an effect of moving the boundary. This is the measurement-lineage discipline inherited from [1]. Coarse-graining can discard information and must be identified separately. Equation (5) is an admissibility rule, not a biological prediction. It does not require identical fitted regression coefficients after conditioning on new variables. Nor does it imply that a different physical clock environment leaves cognition unchanged. It preserves records and definitions across descriptions of the same realization; it does not assert universal causal invariance. Psychometric measurement invariance concerns whether the relation between indicators and a latent construct is comparable across groups, conditions, or occasions [21,22]. Equation (5) instead concerns recovery of an inherited record and its declared measurement specification when the same realization is described at an enlarged analytical boundary. These are complementary requirements: preserving a scoring rule does not establish equal loadings or intercepts across conditions, and an invariant factor model need not recover an earlier trajectory. If latent scores are compared across Test A cells or Test B interfaces, their psychometric comparability requires separate assessment;

equation (5) provides no exemption. The preservation rule is methodological admissibility, not a claim of new mathematics.

5.2 A separate interface-sufficiency hypothesis

An empirical implementation can make the stronger claim that a pre-specified record Z of clock exposure and task interaction is sufficient for transfer across clock environments e . Here Z comprises logged display values and exposure times, clock-reading accuracy, access cost, goal instruction, and motor demands; it does not include the target action or temporal judgment. Let h denote fixed internal-history proxies measured before that target response.

$$p(y_q | h, Z, e = A) = p(y_q | h, Z, e = B). \quad (6)$$

Equation (6) is a deliberately risky hypothesis of the implementation, not a consequence of drawing a boundary. An omitted cue, confidence difference, or incorrect access proxy can violate it. Such a failure defeats the declared sufficiency claim. Adding a new component of Z after the result creates a new version to test on new data. This distinction yields an inference unavailable from an unspecified flow diagram: recover the same inherited history measure under expansion, freeze its meaning, then test whether an interface specification supports cross-environment prediction without outcome-driven redefinition. An ordinary cognitive model that imposes the same restrictions can make the same inference; no exclusive ownership is claimed. The systems contribution is therefore a disciplined comparison across boundaries rather than an additional force acting on temporal experience. The empirical program has two parts: Test A asks whether typed relations constrain prediction within a fixed person-level boundary, with D1a addressing the registered reconstruction pattern and D1b addressing incremental predictive utility; Test B/D2 asks whether an inherited record remains interpretable and sufficient when the analytical/interface boundary is enlarged. D1a, D1b, and D2 test distinct commitments and should not be collapsed into a single success/failure label.

5.3 Methodological neighbors and residual claim

The statistical and transfer tools used downstream of the Level 2 declarations are mature. Informative-hypothesis and order/inequality-constrained methods already encode substantive expectations as equality, order, or inequality restrictions on model parameters and compare those restricted hypotheses [26,27]. The coefficient regions used in D1a and the M3 family therefore are not new mathematics. Their role here is downstream: information lineage, query referent, and any prospectively specified cross-referent transformation determine which relation class is admissible before the experiment-specific region or coefficient restriction is registered.

Current element	Mature neighboring framework	Residual claim here
Rule 3 / D1a / M3 region restrictions	Informative hypotheses; equality/order/inequality-constrained testing [26,27]	The statistical restriction is preceded; Level 2 instead specifies the lineage/referent/transformation rule used prospectively to license it.
Rule 4 / Eq. (5)	Data provenance/traceability [28]; psychometric measurement invariance [21,22]	Recover the inherited record and measurement specification under analytical enlargement of the same realization. Not psychometric invariance and not a new theorem.
Eq. (6) / Test B	Invariant prediction, domain generalization, transportability [29-31]	Implementation-specific conditional predictive invariance for frozen h and Z across two interfaces; not causal transportability or all-environments invariance.

Provenance and measurement comparability are likewise mature. W3C PROV formalizes origins, derivations, and transformations of informational objects [28], while psychometric measurement invariance asks whether indicator-construct relations remain comparable across groups, conditions, or occasions [21,22]. Rule 4 and Equation (5) are narrower: they require recovery of an inherited record and its declared measurement specification when the same realization is redescribed at an enlarged analytical boundary. This is a bookkeeping/admissibility requirement, not psychometric invariance and not a new invariance theorem.

Equation (6) is used here as an implementation-specific conditional predictive invariance claim. Invariant prediction, domain generalization, and transportability already study stability sufficient for prediction or causal transfer across environments [29-31]. Causal transportability addresses stronger questions than the present interface test [29]. Test B instead asks whether, for frozen h and Z across two declared interfaces, interface identity adds practically meaningful predictive information. Its preregistration must therefore define an overlap/common-support criterion for h and Z across $e = A$ and $e = B$, or restrict Equation (6) evaluation to a predeclared common-support region. Unsupported extrapolation does not adjudicate D2 unless separately preregistered.

The residual proposal is limited to the upstream prospective derivation and preservation of admissible relation classes from information lineage + query referent + constrained cross-referent transformation + boundary lineage, with experiment-specific parameter restrictions and transfer tests downstream. A conventional model adopting the same declarations remains empirically equivalent for that implementation. Prospective reuse across genuinely new tasks and boundaries remains open and must be demonstrated rather than assumed.

6. Proposed adversarial tests

6.1 Test A: primary empirical program

The central experiment crosses sequential richness L with represented temporal extent T in a 2×2 design. All target episodes last 90 seconds. Participants see 18 displays at five-second intervals and make a motor response at each display. In the high-richness condition, a task requires retaining and updating a chain of six causally related states. In the low-richness condition, the same number of displays and responses supports three repetitions of a two-state cycle. Scene transitions and sensory transients are yoked. This manipulation concerns task-defined sequential processing, not a measured quantity of consciousness. Independently, a narrative implies that the depicted scenario spans either a day or a decade. Match word count and syntax, and counterbalance narrative identities across conditions. Pilot the materials for comprehension, difficulty, arousal, inferred event boundaries, contextual change, omitted-event inference, causal continuity, and credibility. Temporal shifts in narratives can themselves induce event-model updating [23], internal event structure can bias subjective duration [24], and remembered subevent structure can predict retrospective duration [25]. Therefore neither L nor T is treated as a pure manipulation: event-structure measures and the strongest event-memory competitor are part of the adversarial test.

The primary outcome is an unexpected retrospective judgment of personally experienced laboratory duration, obtained before any question about fictional time. Use this wording: “Compared with the reference task, how long did the time you personally spent completing this task seem to last? Judge your own time doing the task, not the amount of time that passed within the story.” The response is a continuous ratio scale centered on “the same duration,” with shorter and longer anchors; analyze the log ratio under a predeclared response-range rule. The reference is a fixed, previously experienced 90-second task, identified by a neutral task name rather than its duration in seconds. Keep the reference procedure identical across cells. Pilot whether reference familiarization alerts participants to later timing questions, assess such expectations after the primary response, and register how anticipated versus unanticipated judgments will be classified. To preserve retrospective status, use one critical surprise duration trial per participant; allocate the four cells between participants. Separate cohorts provide online passage ratings and prospective numeric estimates. Repeated duration questions would change the task and cannot be described as surprise retrospective timing.

Ask the narrative-extent question separately and later: “How much time passed within the story, from its beginning to its end?” A subsequent referent check asks which interval was judged in the primary question. Register its scoring and exclusion/sensitivity rules before data collection; report the randomized intention-to-treat contrast alongside any comprehension-restricted analysis. Confusion between personal and fictional time invalidates an experiential interpretation rather than providing evidence for the theory. Auxiliary outcomes are proportion correct on a fixed number of serial-order probes, source-specific detail, reconstruction latency, and performance on a matched secondary-memory task. Detail is scored against available information and order probes avoid ceiling and denominator artifacts. A semantic comprehension check asks about represented extent only after the primary response. An independent calibration sample estimates T^* from the scenario's temporal implications. On a planned

subset, source-confusion probes distinguish events shown on screen from events merely implied by the scenario. These measures constrain, but do not directly reveal, process history and representation.

The primary evidence criterion is a conjunction: an independently measured, selective process/representation pattern must support held-out prediction of personally experienced duration that a comparably flexible semantic and event-memory model misses. A positive narrative-extent effect by itself does not meet that criterion. The direct $L^* \rightarrow Y_order$ and $T^* \rightarrow Y_extent$ relations primarily establish manipulation/measurement validity; their status as referent-direct routes is derived prospectively from the source-lineage and query-referent declarations in Section 4.1. The decisive theoretical comparison preserves those relations across competing restricted models and asks whether the additional Rule 3-admissible reconstruction route to personal duration survives adversarial comparison.

Table 2. Test A cells and proposed selective outcomes.

Sequential richness	Short represented extent: one day	Long represented extent: one decade
Low L: repeated two-state cycle	Lower serial-order score; short represented extent. Personal-duration baseline for low L.	Serial-order score should remain equivalent to low-L/day within margin; longer represented extent. D1a predicts a longer personal-duration judgment than low-L/day.
High L: six-state chain	Higher serial-order score than low-L/day; represented extent equivalent within margin. No directional personal-duration contrast with low L is required.	Higher serial-order score than low-L/decade; represented extent equivalent within margin. D1a predicts a longer personal-duration judgment than high-L/day; no directional high-L versus low-L duration contrast is required.

All cells last 90 laboratory seconds. These are proposed contrasts, not observations. Equivalence requires registered margins. The cellwise pattern and coefficient regions contribute to D1a; D1b separately requires the held-out model comparison in Section 6.2.

6.2 Decisive Test A comparisons: D1a and D1b

For this implementation, define three different response channels: Y , the primary personal-duration log ratio; Y_order , source-verified serial-order performance; and Y_extent , represented narrative extent. L^* and T^* are calibrated in independent samples using probes distinct from the held-out target responses. T^* is the same independently calibrated semantic/temporal-extent observable supplied to M2 and M3; it is not an additional hidden predictor available only to M3. No target Y , Y_order , or Y_extent is reused to construct its own predictor. Item features inferred in calibration data must be estimated without access to the confirmatory sample, and uncertainty in those estimates propagates into prediction. Here L labels the randomized high/low sequential-richness condition; it is one implementation of sequential-processing richness, not an operational definition of Γ_B or a measure of the whole process history. The high/low represented-extent condition is abbreviated T in the experimental design and must be distinguished from the general content object $T(u)$. L^* indexes task-defined sequential structure and its process-derived record; the architecture does not require L^* to monotonically lengthen Y . Its primary registered relation is to Y_order , while its coefficient in the personal-duration model remains free. T^* indexes represented temporal extent; its primary registered relation is to Y_extent , and the reconstruction implementation additionally predicts a positive contribution to Y . Working-memory demand, novelty, predictability, causal complexity, retrieval burden, and event structure remain competing explanations for L .

The proposed positive $T^* \rightarrow Y$ effect rests on a functional reconstruction hypothesis: retrieved temporal-extent structure contributes positive evidence of elapsed duration when the personally experienced episode is reconstructed, even when the participant correctly identifies which events belonged to the story. Greater represented extent is therefore predicted to increase Y under this implementation, without requiring a proportional increase in source-verified serial history. This is an additional empirical hypothesis, not a deduction from the typed architecture: effective source separation could prevent that contribution, and a reliably absent or reversed effect would count against the registered positive-effect implementation. The rationale is also compatible with semantic reconstruction, so it does not replace the held-out comparison with M2. Test A separates the registered evidence into two decisions. D1a, the reconstruction-pattern hypothesis, requires three findings together. First, high versus low L improves Y_order at both narrative extents within the piloted non-overload regime, while the T^* cross-relation to Y_order lies

within a predeclared equivalence margin. Second, long versus short represented extent increases Y_{extent} , while the L^* cross-relation to Y_{extent} lies within its own declared margin. Third, represented extent contributes positively to personal-duration reconstruction: long versus short represented extent increases Y within both L levels, including in the held-out narrative family without explicit duration numbers, and the uncertainty set for $\beta_{T,\text{duration}}$ lies in the preregistered positive region rather than the practical-null, opposite-sign, or indeterminate regions. No directional $L^* \rightarrow Y$ effect is required by this implementation. D1b, the architectural-utility hypothesis, asks whether M3-R achieves a practically meaningful held-out gain for Y over unrestricted same-information M2 and over the best preregistered matched duration-routing alternative - including the matched null M3-N - without sacrificing prediction of Y_{order} or Y_{extent} beyond declared tolerances. Nonsignificant cross-effects are not evidence of equivalence. Directional thresholds and equivalence margins are set from pilot precision and the smallest useful difference before confirmatory analysis.

Support for D1a would directly support the registered Level 3 reconstruction-pattern implementation. Support for D1b would additionally show that imposing the prospectively derived routing restrictions earns incremental held-out predictive utility relative to the tested unrestricted and matched alternatives. D1a support does not require D1b support: if M2 matches M3-R while the positive coefficient pattern is supported against matched null/opposite regions, the reconstruction pattern may be supported but the restriction has not earned incremental predictive economy. Conversely, a predictive advantage cannot convert an indeterminate or nonpositive coefficient region into D1a support. Neither decision establishes the Level 1 type distinction as an empirical discovery, biologically separate modules, or every implementation of Boundary Theory. The selective relations are graded structural constraints, not claims that cognition forbids all cross-routing. A competent semantic/event-memory model receives the same T^* and L^* information. Representation shifting Y_{extent} alone is a comprehension result. Representation shifting Y only for literal numerical labels favors anchoring or demand. If a matched duration-routing alternative predicts as well as M3-R, D1b is not supported even if D1a is.

The comparator is specified here because it determines what D1a and D1b can establish. M0 includes clock duration, temporal attention, arousal, and participant/stimulus effects. M1 adds event boundaries, contextual diversity, memory strength, and retrieval delay. M2 is the unrestricted same-information semantic/event-memory competitor: it additionally receives T^* as the

semantic/temporal-extent feature, L^* , generic magnitude anchoring, measured task demand, remaining semantic features, and the same participant and stimulus information supplied to the restricted models. It may estimate separate coefficients for every L^* and T^* path to Y_{order} , Y_{extent} , and Y . M3-R and its matched alternatives receive no extra predictor; they differ only by preregistered constraints on the same coefficient space. The first confirmatory implementation omits $L^* \times T^*$ interactions in all models; adding them creates a separately registered comparison. This formulation deliberately treats T^* as potentially identical to the semantic-duration feature used by M2. No duplicate column and no new latent construct are claimed. The scientific content lies in whether the typed role defines a useful restricted relation space. If formal specification shows that M2 and the restricted family are observationally equivalent, D1b cannot establish incremental predictive economy for the restrictions and the simpler conventional description should be preferred for that purpose; this does not by itself negate D1a or imply that the estimated coefficient pattern is opposite to the registered routing. To separate theory-relevant routing from generic regularization, the decisive restricted controls preserve the direct $L^* \rightarrow Y_{\text{order}}$ and $T^* \rightarrow Y_{\text{extent}}$ manipulation-validation relations and differ only in their preregistered duration-routing commitment. Swapping the direct order/extent routes may still be used as a diagnostic misrouting check, but its expected failure is not treated as evidence for the temporal-reconstruction theory.

Two controls are essential. A non-temporal magnitude label, matched for numerical size, tests generic anchoring. A second stimulus family communicates long versus short extent through relational context without explicit duration numbers, testing transfer beyond label repetition. Train on explicit-label stories and test on the relational family, with family-specific comprehension assessed independently. An effect confined to literal labels and direct represented-extent questions supports semantic comprehension or response bias, not the proposed contribution to temporal experience. The manipulation may still alter attention, expectations, or contextual structure. Therefore a simple double dissociation is necessary evidence for separate measurements but insufficient evidence of a unique architecture. The stronger test is predictive generalization over the established alternatives in Section 7. No sample

size is asserted here: preregistration should use simulation from pilot reliability, a declared smallest relevant effect, and the proposed participant-by-stimulus structure to choose a fixed sample size before confirmatory recruitment.

6.3 Test B: cross-boundary systems-transfer test

Test B carries the separate empirical burden for the boundary-transfer claim. It is not a substitute for D1a or D1b evidence, and neither Test A decision establishes D2. In a four-minute ongoing task, separate direct clock reading, scheduled action, and clock-absent estimates. Cross clock access with a predeclared access cost or deadline incentive; record displayed values, exposure times, checks, reading accuracy, and action timing. The established monitoring and clock-speed studies [14,15] provide the comparison class. Selective action improvement without improved unaided estimation is a consistency result. For the transfer demonstration, fit on one interface and test on a second with matched display trajectories, instructions, and motor requirements. Yoke scheduled exposures during transfer to separate rendering from self-selected monitoring. Freeze h, Z, their scales, and the response definition. Equation (6) predicts no practically meaningful predictive gain from interface identity once those records are supplied. Register equivalence margins for transfer loss and residual interface effects; nonsignificance alone does not demonstrate sufficiency.

Excess transfer loss or a meaningful residual interface effect defeats D2 for the declared implementation. Confidence, visual effort, or another omitted cue may motivate a revised model, but that revision requires new data. The preserved rule of equation (5) cannot rescue a failed equation (6). Details of calibration training and developmental acquisition remain future work.

7. Predictions, reduced models, and failure criteria

7.1 Established consistency requirements

C1: prospective duration, retrospective duration, and passage reports can differ; assess instruction and response timing independently [2,3]. C2: event structure and contextual memory can explain duration variance [4,6,7]. C3: clock-monitoring behavior can differ from deadline accuracy [14,15]. C4: duration represented in memory or language need not match replay or reading time [11,12]. Replicating these results establishes compatibility, not novelty.

7.2 Discriminating implementation hypotheses

D1a: the reconstruction-pattern criterion in Section 6.2 tests the Level 3 within-boundary implementation: primary $L^* \rightarrow Y_{\text{order}}$ and $T^* \rightarrow Y_{\text{extent}}$ relations prospectively classified as referent-direct routes, bounded crossed order/extent paths, and the registered positive $T^* \rightarrow Y$ reconstruction effect instantiated through the prospectively specified Rule 3 transformation, including transfer to the held-out relational narrative family. D1b: the architectural-utility criterion asks whether M3-R yields a practically meaningful held-out gain over both unrestricted M2 and the best matched duration-routing alternative, including M3-N, while preserving the auxiliary-channel tolerances. D1a and D1b are reported independently: M2

matching M3-R can leave D1a supported while D1b is not. Successful generalization can support the usefulness of the Level 2 relational methodology only insofar as the restrictions were generated by the pre-experimental admissibility rules and later prove portable to tasks not used to formulate them; the present manuscript specifies that prospective reusability criterion but does not claim it has already been demonstrated empirically. Neither D1a nor D1b establishes the Level 1 type distinction as an empirical discovery. T^* is not evidence of a new construct by itself. D2: the fixed clock-interface record supports the cross-boundary equivalence claim of equation (6) in Test B. D1a, D1b, and D2 are independent inferential burdens. D3, reserved for later work: an independently manipulated retrieval cue changes the influence of previously encoded content on a temporal response while initial encoding is held fixed. Retrieval success must be measured separately; mere visibility or self-reported awareness is not a sufficient access manipulation. D3 would support a functional-access distinction without proving a phenomenal-access theory.

7.3 The strongest reduced competitor

For Test A, let r in {order, extent, duration} index the three response channels, i participants, and j stimuli. Let X_{ij} contain the common competitor predictors: temporal attention, arousal, event-boundary and contextual-diversity measures, memory strength, retrieval delay, generic magnitude anchoring, task demand, and remaining predeclared semantic features. L^*_{ij} and T^*_{ij} are standardized calibrated observables, with T^* supplied identically to M2 and M3. Both model classes use the same identifiable participant and stimulus effects and the same outcome-specific likelihoods. Let κ_r indicate whether an outcome-specific participant random effect is identifiable from repeated observations in that channel.

$$\eta_{ijr} = \alpha_r + \kappa_r u_{ir} + v_{jr} + X_{ij}^T \gamma_r + \beta_{L,r} L^*_{ij} + \beta_{T,r} T^*_{ij}. \quad (7)$$

$$Y_{ijr} \sim F_r(g_r^{-1}(\eta_{ijr}), \psi_r). \quad (8)$$

For the registered implementation, F_{order} is an item-level binomial model with a logit link for fixed-denominator serial accuracy; Y_{extent} and the log-ratio personal-duration outcome Y use a predeclared continuous likelihood selected from pilot residual diagnostics before confirmatory analysis and then held fixed across models. In the primary Test A model, $\kappa_{\text{order}} = 1$ when repeated order probes identify participant heterogeneity, whereas $\kappa_{\text{extent}} = \kappa_{\text{duration}} = 0$ because each participant supplies one represented-extent response and one critical surprise duration response. Thus no separate outcome-specific participant random intercept is estimated for either one-shot continuous channel. Participant-level measured predictors may remain in X_{ij} . A future multivariate hierarchical model that shares participant effects across outcomes would be a separately specified implementation, not an implicit rescue of equation (7). The same identifiable random-effects structure, feature scaling, missing-data rule, and training folds are used for M2 and all restricted alternatives. No cross-outcome score is formed by summing unlike response units.

M2 is the unrestricted same-information competitor. All six coefficients $\beta_{L,r}$ and $\beta_{T,r}$ are free, subject only to the common regularization selected within training folds. Thus M2 can learn any supported routing of L^* and T^* across order, extent, and personal-duration outputs. It is not forced into a single-clock, single-memory, or single-output account. The restricted family is a subset of that same predictor space. All matched members share four direct/cross-channel constraints: $\beta_{L,\text{order}} \geq \delta_{Lo}$; $\beta_{T,\text{extent}} \geq \delta_{Te}$; $|\beta_{T,\text{order}}| \leq \epsilon_{To}$; and $|\beta_{L,\text{extent}}| \leq \epsilon_{Le}$. These common constraints preserve the manipulation-validation routes rather than using their reversal as the decisive adversarial test. The proposed model, M3-R (represented-extent reconstruction), adds one directional duration restriction: $\beta_{T,\text{duration}} \geq \delta_{TY}$, while $\beta_{L,\text{duration}}$ remains unconstrained in sign and magnitude. The restrictions define a graded hypothesis space rather than a claim that cross-routing is biologically impossible. Thresholds are fixed from pilot precision and smallest-useful-effect criteria before confirmatory outcomes are examined.

Because M3-R is more restricted than M2, better held-out prediction could reflect useful regularization rather than the proposed reconstruction routing. The primary matched control family therefore preserves the same four direct/cross-channel constraints and changes only one duration restriction. M3-N (matched null reconstruction) requires $|\beta_{T,\text{duration}}| \leq \epsilon_{TY}$ while leaving $\beta_{L,\text{duration}}$ free; it is the direct same-complexity scientific null to the proposed represented-extent reconstruction route. M3-O (opposite represented-extent reconstruction) requires $\beta_{T,\text{duration}} \leq -\delta_{TY}$ while leaving $\beta_{L,\text{duration}}$ free. M3-P (process-history reconstruction) requires $\beta_{L,\text{duration}} \geq \delta_{LY}$ while leaving $\beta_{T,\text{duration}}$ free. M3-R, M3-N, M3-O, and M3-P therefore contain the same predictors, the same four common validation constraints, and one additional duration restriction. M3-R and M3-P are not mutually exclusive mechanistic theories: if both duration coefficients are positive, the same data-generating structure can satisfy both regions. M3-P is therefore interpreted only as a matched test of whether M3-R has predictive advantage over a process-restricted architecture; M3-N and M3-O provide the cleaner direct adversaries to the positive T^* duration route. A joint no-cross-query model with both duration coefficients bounded near zero and an additive model with both duration routes positive may be preregistered as substantive secondary alternatives, but because they differ in restriction count they are not used as the primary matched regularization controls. Swapped direct-route models are retained only as diagnostic checks of manipulation validity.

The primary predictive score is held-out log predictive density for Y on participants and the relational narrative family not used to fit the model. Y_{order} and Y_{extent} are scored separately with the same proper log score for their likelihoods and must not deteriorate beyond predeclared tolerances. Tuning occurs only inside training folds. Two inferential outputs are reported and mapped to separate decisions. D1a uses a predeclared region-based uncertainty

rule. The preregistration fixes one uncertainty procedure and its level before confirmatory recruitment. For $\beta_{T,duration}$, positive-region support requires the registered uncertainty set to lie entirely at or above δ_{TY} ; practical-null support requires it to lie entirely within $[-\epsilon_{TY}, +\epsilon_{TY}]$; opposite-sign support requires it to lie entirely at or below $-\delta_{TY}$. Any remaining result -including an uncertainty set that crosses a region boundary or an effect lying in the gap between ϵ_{TY} and δ_{TY} - is indeterminate for D1a. The same logic applies to the other registered directional/equivalence regions. A favorable predictive score cannot convert an indeterminate coefficient pattern into D1a support. D1a is adjudicated only by its registered coefficient, contrast, equivalence, and held-out relational-family generalization criteria. M3-N and M3-O can bear on D1a only through the corresponding predeclared parameter-region or contrast evidence; held-out predictive parity or superiority by those models does not by itself overturn an otherwise supported D1a pattern. D1b asks whether imposing M3-R yields a practically meaningful held-out Y gain over unrestricted M2 and the best preregistered matched duration-routing alternative, with uncertainty intervals, while satisfying the two auxiliary-channel tolerances. If M3-N, M3-O, or M3-P matches or exceeds M3-R on the registered predictive criterion, M3-R has not earned the corresponding predictive advantage and D1b is not supported against that alternative. Because M3-R and M3-P can overlap, their predictive comparison is not interpreted as choosing a unique causal mechanism. If M2 matches M3-R while D1a is otherwise supported, the routing pattern may be present but D1b is not supported because the restriction has not earned incremental predictive economy over its encompassing model. Joint no-cross-query or additive alternatives are interpreted according to their stated hypotheses rather than as matched regularization controls.

For Test B, the models are scored separately from Test A. The frozen h and Z definitions are fit on one interface and evaluated on the second. D2 fails if interface identity provides a practically meaningful residual predictive gain or transfer loss exceeds the registered margin. The preservation rule in equation (5) cannot rescue that empirical failure. The proposal therefore has several failure levels. Failed manipulation checks defeat the experiment's interpretation. Failure to generate the registered routing classes prospectively from the Section 4.1 declarations, or reliance on a cross-referent transformation that is only named rather than operationally/theoretically specified with a discriminating consequence, defeats the claimed prospectively reusable Level 2 interpretation for that route. D1a is not supported if the registered direct/cross-channel constraints fail, if $\beta_{T,duration}$ is practical-null, opposite-sign, or indeterminate under the preregistered region rule, or if the held-out relational-family effect does not generalize as declared. Parameter-region or contrast evidence favoring the M3-N null region or M3-O opposite-sign region therefore counts against D1a; predictive parity with M3-N or M3-O alone does not. D1b is not supported if unrestricted M2 or the best matched duration-routing alternative - including M3-N, M3-O, or M3-P - matches or exceeds M3-R within the predeclared meaningful-gain criterion, even when D1a is supported. M3-P predictive parity removes any predictive advantage over a process-history-restricted architecture without identifying a unique alternative mechanism. A joint no-cross-query model matching or exceeding M3-R would further undermine the incremental predictive case for cross-query reconstruction. Excess interface transfer loss defeats D2. Revised constraints remain new versions requiring new data; an all-encompassing scalar is not required for the framework to lose.

8. Ordinary examples and the altered-state stress test

8.1 Three illustrations

Waiting illustrates why external sparsity need not imply sparse internal processing. Boredom or consequential anticipation can involve repeated monitoring and bodily changes; which of these were encoded and retrieved must be measured. Absorption supplies a contrast: a task may involve many updates while attention to clocks is low. It does not follow that its remembered duration must be short; event structure and retrieval can reverse that outcome. Exercise separates a felt slow interval from a later compressed recollection. These examples motivate measurement combinations and do not validate their proposed causes.

8.2 DMT as hypothesis generation

DMT remains an advanced stress test rather than an initial validation target. Lewis-Healey et al. [19] combine EEG with time-resolved measures of experience, supporting attention to changing experiential dimensions. Such work does not convert a retrospective description into a direct count of sequential conscious transitions. The present manuscript makes no factual claim that people traverse centuries during an acute drug interval. Consider hypothetically a report

that an episode contained an extremely long history. Four explanations remain open: unusually dense processing; altered ordering; compact content implying a long interval; or reconstruction during later reporting. “Timelessness” need not mean either long or short numerical duration. Separate questions about passage, sequence confidence, implied temporal extent, and source-specific detail are needed, and time-resolved tracing itself can be retrospective.

These possibilities are not uniquely psychedelic and should first be studied with ordinary narrative and memory tasks. A drug study would add expectancy, reporting difficulty, sampling disturbance, and source-memory uncertainty. No therapeutic, diagnostic, developmental, or durable recalibration inference follows. Any claim that external-clock feedback changes later K_D requires independent later unaided testing and an appropriate comparison condition.

9. Limitations and future work

The architecture does not establish the existence of separately stored process and representation modules. The distinction is between object types and functional roles, compatible with distributed, overlapping, and dynamically reweighted implementation. Boundary and typing declarations constrain which relations are referent-direct and which require an additional transformation; they do not identify a single anatomical 'thing' or require a rigid controller. A transformation label is not an escape hatch: if its input, target referent, operational/theoretical basis, and discriminating consequence cannot be specified before outcome inspection, that cross-route remains exploratory and cannot count as Level 2 support. The key Level 2 risk is prospective reuse: the present article specifies a procedure intended to be carried to new tasks, but does not demonstrate cross-task reusability. If the information-lineage, query-referent, transformation, and boundary rules cannot classify relations in a new task before its outcome pattern is inspected, then the typing has collapsed into experiment-specific labeling rather than a reusable systems methodology. Neither the full process history nor phenomenal experience is directly observable. Proxy reliability and causal ambiguity therefore limit all proposed tests.

The boundary rule is methodological. Its empirical extension depends on the independently specified interface-sufficiency hypothesis. An existing cognitive model with identical preservation, lineage, referent, transformation, and transfer commitments would be equivalent for this purpose; no exclusive statistical machinery is claimed. The distinction from generic domain-informed constrained modeling is therefore procedural and representational rather than algebraic: the admissibility map is declared before coefficient restrictions and must remain portable to new tasks and enlarged boundaries. Similarly, established work already distinguishes replay from represented duration and internal estimation from external monitoring. The present review is targeted rather than exhaustive, so novelty cannot be guaranteed by omission from this bibliography. The most consequential unresolved questions are whether D1a's represented-extent reconstruction pattern occupies its preregistered positive region and generalizes to the relational narrative family, whether D1b's imposed restrictions add predictive utility beyond a competent same-information semantic/event-memory model and matched alternatives, and whether the Level 2 procedure can be carried prospectively to tasks not used to construct Test A. T* is intentionally not protected as a novel latent construct. The strongest matched null is M3-N, which preserves all common order/extent constraints but bounds the represented-extent contribution to personal duration near zero while leaving the L* duration route free. If the registered parameter-region or contrast evidence favors the M3-N null region or M3-O opposite-sign region, D1a fails. If M3-N or M3-O instead merely predicts as well as M3-R while D1a's coefficient/contrast/generalization criteria are satisfied, D1a remains separately supported but D1b has not established predictive advantage over that matched alternative. If M3-P generalizes as well as M3-R, D1b has no predictive advantage over a process-history-restricted architecture. If M2 matches M3-R while D1a is otherwise supported, the restriction has not earned incremental predictive economy even though the coefficient pattern need not be false. If a new task requires the routing rules themselves to be invented after its response structure is observed, Level 2 has not earned independent theoretical content beyond constrained-model preregistration. A familiar dissociation relabeled with L and T would not suffice.

Clock-calibration biology, developmental acquisition, and retraining are future work. Their investigation should first establish which function changed: clock recognition, unaided mapping, monitoring, action selection, or passage experience. No developmental causal pathway, disorder-specific explanation, or treatment mechanism is advanced. Cultural differences in clock use also limit generalization from the adult laboratory tasks proposed here.

10. Conclusion

A useful systems account of temporal experience and judgment should preserve the distinction between the processing a system undergoes and the temporal extent represented by its contents, without pretending that either object has a single rigid behavioral consequence. The present proposal keeps three inferential levels separate: Level 1 supplies conceptual object typing; Level 2 supplies a prospective systems methodology for generating and preserving admissible relations; and Level 3 supplies falsifiable registered implementations. Level 2 begins with information lineage, query referent, transformation specification, and boundary-lineage preservation. Referent-direct relations require no referent-changing operation, whereas a cross-referent route is admissible only when the transformation's inputs, target, operational or theoretical basis, and a discriminating empirical or measurement consequence are predeclared; naming an explanatory operation alone is insufficient.

Experiment-specific coefficient restrictions instantiate rather than create that architecture. Test A then separates D1a, support for the registered positive represented-extent reconstruction pattern, from D1b, incremental held-out predictive utility of imposing the M3-R restriction relative to unrestricted and matched alternatives. D1a may be supported even when D1b is not, and neither establishes Level 1 as an empirical discovery. Test B/D2 separately asks whether inherited measurement meaning and interface sufficiency survive analytical enlargement. A conventional model adopting the same declarations is empirically

equivalent for a given implementation. The framework therefore specifies a prospective reusability criterion rather than claiming that cross-task reuse has already been demonstrated; it earns value only if those Level 2 rules can be carried to new tasks and expose predictive or transfer structure that established accounts do not capture as well.

Data availability

No empirical data were generated or analyzed. The experiments and model comparisons are proposed research designs.

Generative AI Disclosure

OpenAI ChatGPT was used as a research-assistance tool during the 2026 research and manuscript-development process. Uses included conceptual formalization; literature discovery and organization; adversarial critique and stress-testing; drafting and editing; notation refinement; simulated peer-review workflows; and submission-package preparation. The human author originated and directed the research questions and analytical program, adjudicated model outputs, retained intellectual control and final scientific judgment, and reviewed or verified cited sources and relevant factual claims. AI is not an author and generated no empirical data or research dataset. OpenAI ChatGPT (GPT-5.6 Sol) was used during final submission-package preparation.

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